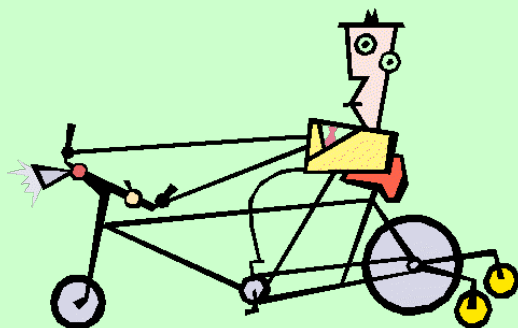


Recumbent Bikes

By [John Andersen](#)

You've probably seen them, those odd shaped bicycles having more in common with lawn furniture than typical bicycles. Perhaps you're wondering what they are all



about and why are there starting to be more of them around. You probably would like to try riding one some time, and wonder if they are easy to ride. Or perhaps you just decided to see if this article would explain just what kind of madness would make a healthy looking bicyclist climb aboard a pedal driven lawn chair.

I'm referring to Recumbent Bicycles of course, those sit-down bikes, which are pedaled with the cyclist's feet out in front, while the rider is seemingly relaxing in a mesh seat looking for all the world like it is no work at all.

In this article I'm going to attempt to answer your questions in a balanced way, because there is a lot of hyperbole both Pro and Con. Let me start right off by saying I use a recumbent and three diamond-frame (regular) bikes as well as a tandem. I like them all, and have spent enough time on each of the types to know the faults and well as the strong points.

What are these things

Recumbent (meaning seated) bikes have been around for quite a while, but have never garnered a significant share of the market. The last I heard, they had just slightly more

sales than Tandems, and you know how rare those are. Yet lately you are starting to see them in increasing numbers.

Recumbents or "Bents" as they are often called in the US and Canada (but seldom in Britain due to another connotation of the word) started out as a tinker's project. Often they were, and some still are, assembled in garages out of pieces and parts of cannibalized bikes by guys with a welding torch in their hand and a gleam in their eye. Today, there are a couple dozen manufactures of quality production recumbents in the US, Europe, Australia, and the Orient.

Out of this environment came a few bikes with very nice handling characteristics, a fair bit of speed, and a great deal of comfort. Comfort seems always to be measured against the alleged horrors of the regular bike seat, usually with dramatic exaggeration thrown in for good measure.

Make no mistake about it, the garage mechanic's bike and the production recumbents which descended from them are comfortable. And yes, they are easy to ride, although in your first five minutes on one you may think me daft for saying so.

Recumbent bikes are ***DIFFERENT***. They ride differently, they steer differently, starting and stopping is different, and climbing hills requires a different approach.

By and large they are quite fun to ride. The "Recumbent Grin" is the most noticeable attribute of a new bent pilot. (That term "pilot" comes up often among recumbent riders because the feeling of banking into turns as you swoop around corners has much in common with feelings you get while flying in small planes).

Recumbent Variety

There are a lot of different recumbent shapes and sizes. Once you break the "rules" and deviate from the diamond frame design there is little to hold you back from trying new designs and new shapes.

The two most common classifications of recumbents are based on where the front wheel is in relation to the rest of the bike. These are short wheelbase and Long wheelbase. There is arguably a third category called compact long wheelbase, but the difference between this and the other categories is blurred and indistinct.

Short

Short, or short wheel based bikes usually have the front wheel about at the rider's knees, while the cranks are way

out in front. These tend to be speedy bikes. Sometimes this yields a harsher ride as you are sitting almost on top of the front wheel.

Long

Long Wheel Based bikes have the front wheel out in front of the cranks like a conventional bike. Such bikes have larger turning radiuses, and some of the smoothest rides of any bike. These bikes also tend to be the lowest bikes, often having the seat less than a foot above the pavement.

Steering Arrangements

Within the three above general categories of bike geometry, there are two common arrangements of the steering mechanism. These varieties each have their own strong points and near fanatical adherents. Both work, both are easy to learn to use.

Above Seat Steering

Above seat steering (ASS) recumbent have handlebars above the seat, usually about chest high. This arrangement is often said to be more comfortable for the beginner than is under seat steering. Above seat handlebars also lead to a more aerodynamic configuration on the bike as your arms are in front of you and therefore do not present an additional wind target.

Under Seat Steering

Under Seat Steering (USS) has the steering assembly under the riders seat and handlebars protruding out from under either side of the seat. This usually entails some sort of linkage to the front wheel, except in the case of short wheel based machines, which have the front wheel very close to the rider.

USS is quite comfortable once you are use to it. There is nothing in front of you. Some riders find this arrangement makes it easier to get on and off, due to not having to duck under the bars.

Not Just Bicycles

Not all recumbents are bicycles. Some are Trikes.

Trikes have a

special appeal to the touring crowd as you are never at a loss for a place to

rest, and the carrying capacity is great. Plus, your full loaded bike does not try to fall over the minute you turn your back. Trikes also appeal to those users with balance problems, salvaging an otherwise lost love of cycling.



There is no learning curve, you can ride one instantly. You can also climb very steep hills because Trikes have incredible gearing (some as many as 72 gears), and also because you do not have to maintain enough speed to balance as you would on a two wheel vehicle. You can set the brake, and stop to rest anywhere without getting off the bike.

Most commercial recumbent trike designs are of the "tadpole" design with two front wheels and one rear wheel. The trike pictured here is a [Greenspeed](#) Touring Trike manufactured in Australia but sold the world over, available either as completed bikes or kits to build your own.

Reasons for Recumbents

There are several good reasons to ride a recumbent bicycle. I will cover these below in greater detail.

But first I would like to mention the fact that often people choose a recumbent after they have many years of using other bikes, and after they have reached an age where they are no longer socially insecure. In prior years, recumbent riders were overwhelmingly middle aged. Lately you are starting to see more young riders, but still virtually no teenage riders. This may well change, as recumbents become more common, not to mention cheaper.

Comfort

One of the most frequently cited reasons for riding a recumbent bike is the comfortable seat. Many people find the having weight on their arms or wrists is unbearable for longer periods with advancing age. Others just never seem to get used to a typical bike seat and find the lawn chair like



seats of recumbents more comfortable.

Make no mistake about it recumbents are very comfortable. However, many recumbent enthusiasts go way overboard in denigrating the typical bike seat, claiming they would never go back to that "horrible" seat. However, the upright's seat would not seem so uncomfortable if the rider rode more frequently.

Long tours are especially comfortable on some of the long wheel based recumbents. You can ride all day and not have any sore spots. Also, as you sit in a more natural posture, your neck does not get sore. and you see more of your surroundings.

Stress Injuries

Recumbents are often a way whereby people with repetitive stress injuries to the wrist, neck, or back can continue to ride bikes. Quite often such injuries make it impossible to bear any weight on ones hands, as is necessary when leaning on the handlebars. Injuries to the back or neck can also keep people off a regular bike, but seldom present a problem for recumbent bikes.

Cycling is still possible with many such injuries if the position on the bike is changed. The upright seating posture of most recumbents is more like sitting in a chair, and reduces stress on arms and neck. Offsetting this, often there is additional stress is placed on knees, especially among new recumbent riders. We will cover this below.

Speed

Recumbents hold all [human powered speed records](#). Period! The world's cycling organizations, or even national cycling organizations such as the USCF recognize none of these records. These organizations have decided that the diamond frame bike (traditional road bike) is the only device they will admit to the record books. Recumbent enthusiasts insist that this is because they know that recumbents are faster and would take all records if given a chance.

Of course that's true, but the winners of recumbent records would be the same world-class cyclists who race in the Tour De France. So a recumbent isn't going to make you a world record holder. You still need world class legs and lungs. Therefore the argument that the USCF "fears" recumbents seems rather weak in that the same super athletes (USCF members for the most part) would still be winning.

The truth is that every sport has the right, in fact the duty, to limit the equipment that may be used. Equipment that lends a tremendous advantage does not reflect the merits of the

athlete. Sport, after all, is supposed to be a competition of strength and skill among human beings. Competitors should rightly be limited to similar equipment. There was a time, sadly long past, when the Tour provided the same bikes to all competitors.

None of the above does anything to diminish the fact that most recumbents are fast bikes. This is due predominantly to the better aerodynamics of a recumbent bicycle. The cyclist is usually seated lower, (wind speed near the ground is usually less than found just a couple feet higher up), and the seating position on most recumbents provides a smaller wind target than most upright bikes.

This translates into measurably faster speeds. Many cyclists who ride both types of bike (diamond frame as well as recumbents) report 3 to 8 mph faster average speeds on the recumbent bikes.

Add to this the low seating position, and the effect is not unlike being in a low slung sports car. Sitting low to the ground just seems faster than the same speed at a higher elevation.

Safety

Virtually all falls from a recumbent dump you unceremoniously on your butt next to your bike. It is almost impossible to go "over the bars" and land on your head. Very few crashes result in extensive road rash as is often experienced in falls from higher bikes. Crashes with Trikes are virtually unheard of.

It is possible to slide out in a tight turn, but if you do so on a recumbent, you will usually have a lot shorter distance to fall than you would from a regular bike.

Because of the novelty factor (see below) recumbents often get more attention and consideration from motorists.

View

When was the last time you were able to ride along the road looking up at soaring eagles rather than down at the pavement ahead of you? On a recumbent bike the view is wonderful, and you see things you never notice on an upright bike. After a long ride, your neck is not tired from holding your head up all day.

Fun

‘Bents are just plain fun to ride. It’s a whole different experience. Among recumbent enthusiasts there is

something known as the "recumbent grin". You will see it on the face of every new recumbent rider just seconds after they master the steering and balance. You bank into corners like a jet fighter pilot. You are riding low and fast like an Italian roadster on a mountain road. You can stop and rest still sitting on your bike.

Novelty

There is also a certain novelty factor in riding a recumbent, which is diminishing with each year as more and more of them are seen on the road. Still, hardly a day goes by without some 10 year old blurting out "Cool Bike, dude!" as you ride by. Motorists point you out to car-mates. Some folks just stop and stare. Everyone smiles and waves.

Recumbent riders end up answering a lot of questions. Many people want to know if they are hard to ride, how much they cost, how do you steer that thing (especially the USS bikes), etc. While paused at a stop sign one day, a guy in a pickup pulled up in the next lane, rolled down his window and asked me "Did your wife put that thing together for you"?

Disabilities

Finally, there are other types of Recumbents that are designed for people who can't use their legs. These are Handcycles, and are either "pedaled" with hand cranks or by hand-on-wheel (like wheel chairs on steroids). These tend to be Trikes, and some are amazingly fast. There are several companies specializing in these bikes, such as [Greenspeed](#), and [Varna](#), and others. There are [sites that sponsor rides and races](#).

Reasons Against Recumbents

In spite of all the positive aspects of recumbent bicycles there are some drawbacks. Not all 'bents have every one of these disadvantages, but most of them have at least one. Some are simply perceptions, others are based in fact.

Weight

Almost every recumbent weighs more than an equivalently priced road bike. There are certain exceptions, you can buy a 21-pound recumbent, but it will end up costing you much more than the 21-pound diamond frame upright bicycle.

Recumbents sometimes weigh as much as 5 pounds more than the same priced traditional bike. On flat ground, the recumbent will be faster, so who cares about weight? Read on.

Hill Climbing

Hill climbing on a recumbent is different than on a diamond frame bike. First you can't stand up and charge up the hill. You have to adapt to a different style of climbing, namely spinning high RPMs on the pedals in a low gear. (In fact, recumbents generally improve your spinning skills and make you a better all round cyclist even on your upright).

Because recumbents are generally heavier, climbing problems are exacerbated by weight. In fact one school of thought is that weight is the only reason recumbents are said to climb slower.

New recumbent riders typically climb slowly until they develop "recumbent legs". This can take one to three months or more depending on how often you ride. There is something about the seating position that requires different muscle development. I'm not sure if it is because your major leg muscles are hanging from your bones in a different maner or because you are sitting on your "gluteus". In either case, every 'bent rider I've ever met agrees that it takes different muscles to get back up to the same level of climbing performance as on your upright bike. On the flats, this is offset immediately by the lower drag, but the hill-climbing disadvantage generally takes much longer to overcome.

However not every recumbent rider agrees that 'bents climb slower. Some recumbent advocates don't ride upright bikes much and have become specialized toward the recumbent. Because of this they can't climb well on an upright and therefore claim that recumbents climb better than upright bikes. But those riders who switch back and forth between recumbents and uprights generally agree that recumbents do not climb as fast as upright bikes.

Knee Stress

Above, I mentioned that you couldn't get up out of the saddle to climb hills. However, just because you climb sitting down does not imply less physical stress on your legs.

Because your back is against a firm seat back, you are easily able to push harder on a recumbent than you could on an upright bike. On the upright you can't push much more than your weight. As soon as you do, your body rises up, and the effort is wasted lifting your body rather than turning the crank.

On a recumbent, you can push against the seat back. You can therefore put more pressure on the pedals - and your

knees. New recumbent riders frequently complain about sore knees. Veteran 'bent riders all reply in unison "Gear Down, Spin Faster".

The advice is simple. Downshift to an easier gear, and spin faster rather than pushing so hard. You achieve the same speed, and it's easier on the knees. Your endurance will be enhanced as well. It takes time to get used to it. A cadence computer helps.

Lower Position and Visibility fears

New recumbent riders generally find themselves sitting much lower than they did on their diamond frame bike. This can be unnerving in traffic, as it is harder to see around or over cars.

On most recumbents you are sitting with your head at the same level as the driver of a modern sedan. This is not that low, but seems like it.

This often leads to fears that you will not be seen in traffic.

The car immediately behind you will see you just fine. The one behind that one (e.g. second car back) may not see you as well if your bike is low as compared to an upright bike. This has some ramifications when you are riding beside a steady stream of higher speed overtaking traffic. Cars may right hook you (turn into a driveway directly across your path).

Most of this is all in the head of the cyclist. The new lower position simply rekindles old fears of being hit from behind. Once you get used to it the lower position is not a problem. Defensive tactics used to avoid the right hook work as well on recumbents as on uprights.

Difficulty of Rearview

Because you have your back leaning against a seat back, seeing behind you can be a problem. It requires greater effort to look back, and may involve leaning forward in your seat.

Because of this, most recumbent riders gravitate toward mirrors, either mounted on the bike or helmet/eyeglass mirrors. The helmet/eyeglass mounted mirrors have the advantage of no blind spots.

Less Carrying Capacity

Many recumbent designs, most of them actually, have small front wheels. This makes it difficult to hang front panniers

on the bike. Most recumbents can accommodate rear panniers just fine.

The rear tire of most recumbents is already carrying more weight than the front. This is true in most, but not all, designs. Some bikes are set up so that the rear wheel carries 70 % of the weight. You compound this when you add the weight of loaded panniers. This calls for a high-pressure tire in the rear. Still the bike will be very tail-heavy. This affects handling, but usually presents no serious problem.

To offset the inability (or simply the uselessness due to small size) of front panniers, recumbents offer a large area behind the seat that can be used to stow a lot of gear. Often this area can be enclosed in a cloth-and-wire frame or rigid structure called a **tail cone**. This makes an excellent carrying area. Commercial versions are available, and home built tail cones are constantly being talked up on the Internet.

Tail cones generally improve the aerodynamics of the bike, or at the very least do it no harm, and provide a large enclosed space for carrying a significant load. They do add to the weight, but the weight gain is almost always offset by improved aerodynamics that make for a net gain in speed. Some claims for a 30% drag reduction are made (on top of the already lower drag of the bike).

Getting Used to a Recumbent

Riding a recumbent is different enough from a diamond frame bike that there is usually a period of adjustment where you master the skills.

This period of adjustment varies in length for different people and different skills.

The good news is that most people can transition to a recumbent in seconds. My first recumbent ride was at the Ryan factory. Dick Ryan put me on his own personal bike and said he would run along side with one hand on the seat back to prevent a crash. After six paces, he gave up as it was clear that I would have no problem steering and balancing.

Real competency came within a week or two. I had to plan starts and stops, and major turns for a few days. I had to train my legs. But I could ride instantly.

There are some differences in handling that you might initially be aware of. Let me cover a few here.

Balancing

A bicycle is a steer-balanced vehicle. You have to steer to balance. On upright bikes, you can also lean to steer, and lean to make small balance adjustments.

On a recumbent, your back is against a seat back. This means you can not lean over like you would on a diamond frame bike. It's possible to do so, but you can't do it quickly enough to be of much value.

Therefore, recumbent riders rely to a larger degree on small steering movements to fine tune balance. This comes naturally, as this is one of the fundamental elements of riding a bike, whether you are conscious of it or not.

You can't simply lean to the left to counteract a slow fall to the right on a recumbent as you would on an upright. As you start falling to the right, you must turn to the right. This brings the bike back under you and you are once more balanced.

Turning

At first, your rides on a recumbent will be sort of deliberate. You will be conscious of planning turns, stops and starts. This lasts about a week, varying somewhat depending on the type of recumbent, and the amount you ride.

Hint: Turns are easier once you understand the concept of counter-steering. Counter steering means that when you want to turn right, steer left. This causes the bike to lean to the right (because the front wheel has moved left of the center of gravity). A lean to the right is followed inevitably by a turn to the right. Counter steering is usually more effective on a recumbent than and upright bike. (But it's used on uprights too).

Pedaling

As stated above, gear down and spin. Even if it means going slower for a while. It's better for your knees, and you will be faster in the end.

Starting Out

New recumbent pilots tend to be wobbly when starting out from a standing stop. This is because you can't jump on the bike and mash a pedal before you are even seated. Some recumbents are configured so that you can push off with your foot, others are just too low for this to work. Doing a start smoothly requires that you have a pedal cocked and your first stroke must be fairly aggressive. Wobbly starts add to the "geeky" look of the bike, and are something most recumbent riders aim to avoid. Often this is just an

appearance of wobbly-ness as the rider makes rapid and fairly large steering inputs, but the actual track of the bike is often just as straight as an upright bike.

Note that the appearance of wobbly starts are also the reason, rightly or wrongly, that recumbents are banned from many mass-start bike events.

Accessories

With recumbents, some of the bike clothes and accessories you thought you had to have are no longer necessary. The first to go are the padded shorts. Not necessary. In fact bike shorts are not even necessary or particularly desirable. Regular shorts work fine. Jerseys with pockets in back or water bottle holsters are not needed. Cycling gloves are not really needed because you will not have any weight on your hands, nor will you be reaching down to wipe glass shards from the tire. You generally can't reach the tire.

Most recumbent riders gravitate to clipless pedals, which prevent your foot from falling off the pedal when you are tired, and improve your cadence.

You will likely want a mirror, and as stated elsewhere, helmet or eyeglass mounted mirrors work best.

Are Recumbents Better?

Are recumbent bicycles better than the traditional diamond frame? The answer is not clear because the question is too broad.

They are better in some ways and not as capable in others.

Recumbents are Different. The difference makes for a lot of fun. They generally are faster on the road than a diamond frame bicycle. Especially on the flats, where they often enjoy a 30% drag reduction over a the upright. This drag reduction makes for easier days on long rides. Coupled with the greater ability to push against the seat back, this drag reduction can yield explosive acceleration and sustained high speed runs.

Recumbents make excellent long distance touring cycles, once you solve the carrying capacity issues with custom panniers etc.

Since you can't bunny hop on a bent, they are less well suited to trail riding than are upright bikes. Full suspension recumbents are available and are fairly capable in rough road conditions. There are some riders who are able to negotiate trails of moderate to intermediate difficulty on

recumbents, but by and large you do not find them on single track.

Personally, I find that switching between my recumbent and my diamond frame bikes makes me a stronger cyclist. I exercise slightly different muscles, and different cadences. Variety is a **Good Thing**^(tm) and one can never be too rich or too famous or have too many bikes.